



Tecnológico de Monterrey
Interpartial Exam B
2025-Feb-May

Grade

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Course: Algebraic and Transcendental Functions

ID-number: _____

Professor: Fco. Javier Vazquez Tavares

Date: 27/01/2025

Last Names and Names: _____

Group: _____

"In accordance with Tec de Monterrey Student Cod of Honor, in this exam I agree to conduct myself guided by academic honesty and not to use unauthorized or illegal means to do so."

Signature: _____

I Read the following questions and identify the option that best answers each, then write its letter on the line (5 points each)

1. (5 points) _____ Select the option with a quadratic function:

- (a) $(a/b)x + c$ (b) $(a_1/b_1)x + (a_2/b_2)x^2$ (c) $(a/b)x^3 + c$ (d) $(a_n/b_n)x^{1/2}$

2. (5 points) _____ Select the correct domain, such that $f(x) \in \mathbb{R}$, where $f(x) := x^{1/n}$

- (a) $x \in (-\infty, \infty)$ (b) $x \in [0, \infty)$ (c) $x \in (-\infty, 0]$ (d) $x \in [-1, 1]$

3. (5 points) _____ Select the option with the range of the function $f(x) := \text{abs}(x)$ for $x \in (-\infty, \infty)$

- (a) $f(x) \in (-\infty, \infty)$ (b) $f(x) \in [0, \infty)$ (c) $f(x) \in (-\infty, 0]$ (d) $f(x) \in [-1, 1]$

4. (5 points) _____ Select the correct type of function of $f(x) := x^{1/n}$, where $n \in \mathbb{Z}$ (x belongs to the set of integer numbers)

- (a) Quadratic (b) Cubic (c) Exponential (d) Rational

5. (5 points) _____ Select the alternative mathematical notation to express $g[f(x)]$

- (a) $f \circ g$ (b) $g \circ f$ (c) $f \rightarrow g$ (d) $f(g)$

6. (5 points) _____ Select the option of the function that map all the domain into the positive real numbers.

- (a) $f(x) := x^3$ (b) $f(x) := \text{abs}(x)x$ (c) $f(x) := \text{sgn}(x)x$ (d) $f(x) := -x^2$

II Solve the following exercises in an orderly and clear manner. **Answers without procedure will not be considered.** Underline or frame your final answer.

1. (12 points) Tacking into account the following functions,

$$f(x) := \frac{a}{b}x + c, \quad g(x) := x + h,$$

express the algebraic representation of

$$H(x) := \frac{(f \circ g)(x) - f(x)}{h}.$$

2. (12 points) Tacking into account the following functions,

$$f(x) := \frac{a}{b}x, \quad g(x) := \frac{c}{d}x^2,$$

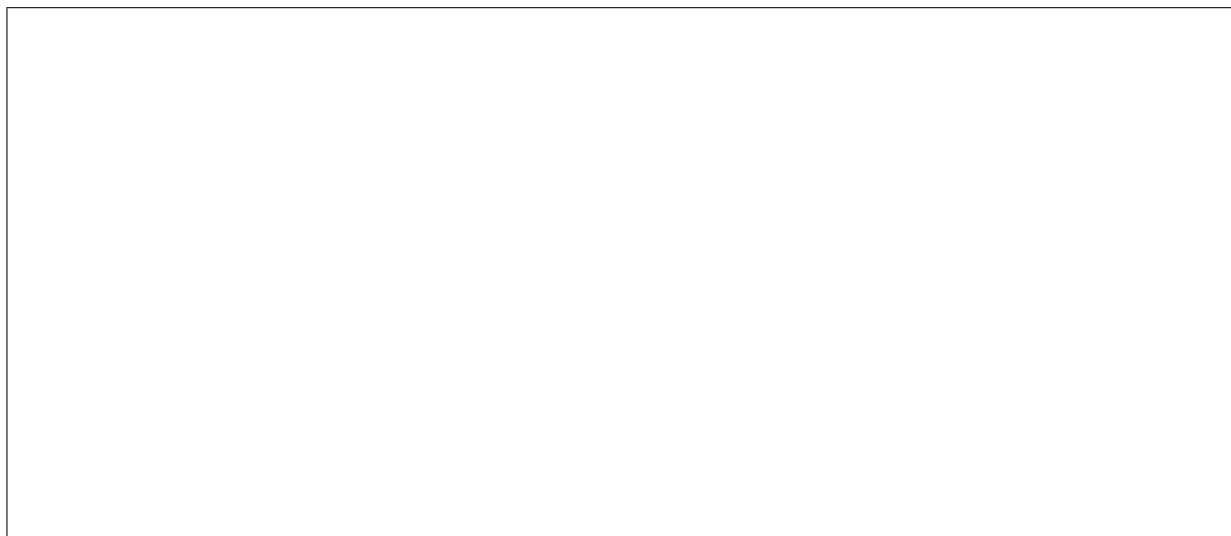
represent the algebraic expression of,

$$\frac{f(x)}{g(x)}, \quad \frac{g(x)}{f(x)}.$$

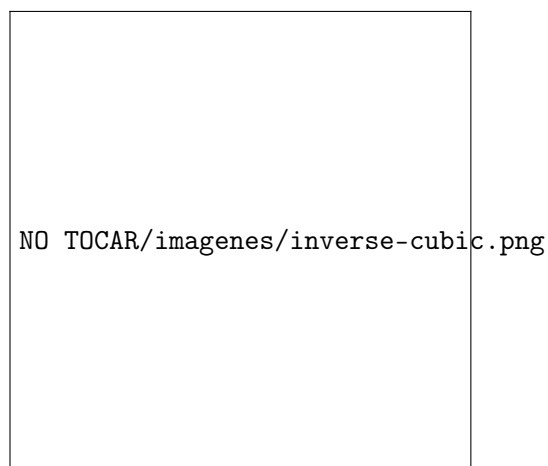
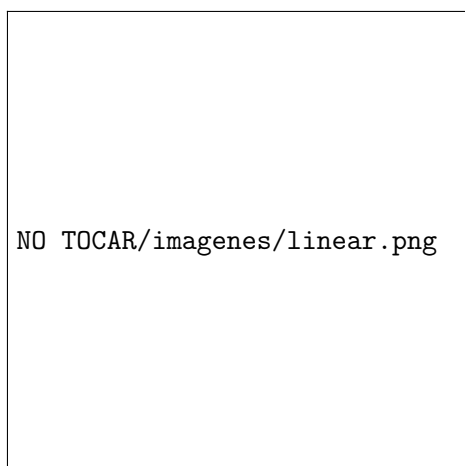
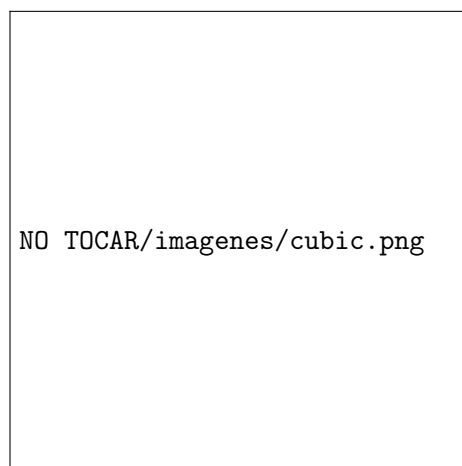
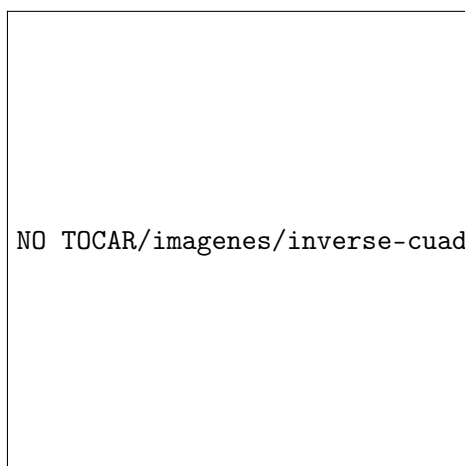
Finally, write the main difference between them.

3. (12 points) Use the verbal representation of the following functions,

(a) $f(x) := ax + c$ (b) $f(x) := (a^2 + b^2)^{1/2}$ (c) $f(x) := (x - 1)^{1/2}, x \in (1, 10]$



4. (12 points) Write a general algebraic expression for each graph,



5. (12 points) Graph a sketch of the following function,

$$f(x) := \begin{cases} x^2 & x \in [0, \infty) \\ -x^2 & x \in (-\infty, 0] \end{cases}$$

NO TOCAR/imagenes/cartesian-plane.png

6. (10 points) Express the domain and range of the following functions,

(a) $f(x) := (x + 1)^{1/2}$ (b) $g(x) := 1/x^2$